Dereq Quinn - Health Data Insights - Take Home Assessment

Date	@February 21, 2026
Document Type	Assessment

Overview

Build a consumer “personal health record + insights” web app using the Metriport Medical API. Users can enter their information, retrieve their records, and explore it through an interactive UI, and use a safe, cited AI chat grounded in their data.

Time box: 20–35 hours. Stop at ~35 hours; write “what I’d do next.”

Docs:

- Quickstart: https://docs.metriport.com/medical-api/getting-started/quickstart
- FHIR overview: https://docs.metriport.com/medical-api/fhir/overview
- Medical Record Summary: https://docs.metriport.com/medical-api/handling-data/medical-record-summary

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What to build

Minimum (required)

UI Flow (Feel free to change)

- **Sign in** (any method) and enforce user isolation.
- **Consent screen before retrieval** (explicit acknowledgment): what data is fetched, how it’s used (insights/chat), delete controls, and (if applicable)

external LLM data flow.

- **Connect Sandbox Record** screen: pick a sandbox persona and link it to the signed-in user.
- Trigger Metriport **Network Query** (async) to pull data and track the status in the UI for the user, store the data for later processing.
- Example UI flow: Sign in → 2) Consent → 3) Select sandbox persona → 4) Retrieval progress screen → 5) Dashboard ready → 6) Timeline drilldown → 7) Deep Insight → 8) Chat with citations → 9) Delete my data

Core UX

Implement these screens:

- **Home/Dashboard:** connected record, last updated, status (ready/running/partial/failed), and 3–6 summary cards (counts, active meds, etc.). CTAs to Timeline, Deep Dive, Chat.
- **Timeline:** filter by category (encounters/conditions/meds/immunizations/labs), time range control, and an event details drawer/modal.
- **Medications view:** grouped list + search/filter + details drawer.
- **Immunizations view:** grouped list + details drawer.
- **Settings:** Disconnect, **Delete my data**, and “AI mode” toggle.

Citations UX requirement: anywhere you show clinical facts, provide sources to back them up.

Insight features

Build an insight feature to help the user gain value from their data in some way.

Medication deep dive :

- computed insights (not just counts/sorting), e.g., likely duplicates, overlaps, changes over time
- a timeline/summary visualization of medication history
- a plain-language explanation of your heuristic + limitations

- drill-down that traces each insight back to underlying health data.

Secure health chat

- Chat UI that answers questions using RAG over the user's stored record
- Every answer must include citations to underlying resources.
- Must refuse medical advice ("should I...") and provide grounded facts only.
- Must address prompt injection: treat user input and record text as untrusted; document mitigations.

Tests (minimum)

- A few unit tests (e.g., parsing/enrichment logic)
- One integration or e2e "happy path" test (lightweight OK)

STRETCH (only if time remains)

- More robust e2e coverage
- Medical Record Summary view
- Immunization and Lab Deep-Dive insight feature
- Mobile responsive

Deliverables / Submission

Submit:

1. Github Repo link (or zip) with clean commits
2. README including:
    - setup instructions + `.env.example`
    - architecture diagram
    - data model summary
    - threat model + retention/deletion
    - product rationale: user problem, why the deep insight helps

    - “how this scales to millions in the US” (high level + compliance considerations)
    - tradeoffs + what you’d do next
    - AI Code set up - what AI tools are you using, and how are you using them to build
3. Tests runnable via documented commands
4. Max 15minute video including a product demo, development process and architecture, and your thought process and decisions made.

Acceptance Checklist

- [ ] Runs locally from README + `.env.example`
- [ ] Auth with user isolation
- [ ] Consent screen before data retrieval
- [ ] Sandbox persona connect + async retrieval with status tracking
- [ ] FHIR data stored and used downstream
- [ ] Dashboard, Timeline, Medications, Immunizations, Settings — all with citations
- [ ] Deep Insight: computed insights + explanation + drill-down to source
- [ ] RAG chat with citations, advice refusal, and injection mitigations
- [ ] Delete my data works end-to-end
- [ ] Unit tests + 1 integration/e2e test
- [ ] README covers architecture, data model, threat model, rationale, scaling, tradeoffs, AI tooling
- [ ] ≤15 min walkthrough video (product demo, architecture, decisions)

Evaluation Criteria

Senior Full Stack Engineer — we’re evaluating the ability to independently ship a production-quality product end-to-end.

- **Product Thinking & UX** — Is the app intuitive and polished? Does it handle edge cases and failure states gracefully?
- **Architecture & Technical Depth** — Sound system design, data modeling, async pipeline, RAG implementation, and security at the architecture level.
- **Code Quality & Engineering** — Clean code, follows best practices, scalable with agentic development, meaningful tests, well-reasoned tooling choices.
- **Insight Quality** — Are computed insights genuinely useful (not trivial), clearly explained, and traceable to source data?
- **Communication & Senior Judgment** — Clear README narrative, tradeoffs, senior-level scaling thinking, and ability to scope and ship under a time constraint.

Submission

Please submit the following by 5pm PST 2/27/2026:

- A GitHub repository or a `.zip` file with all materials
- Video walking through the product and your thought process. (Hit the points in the evaluation criteria)
- Share the repo / zip file with albright@coretsu.com